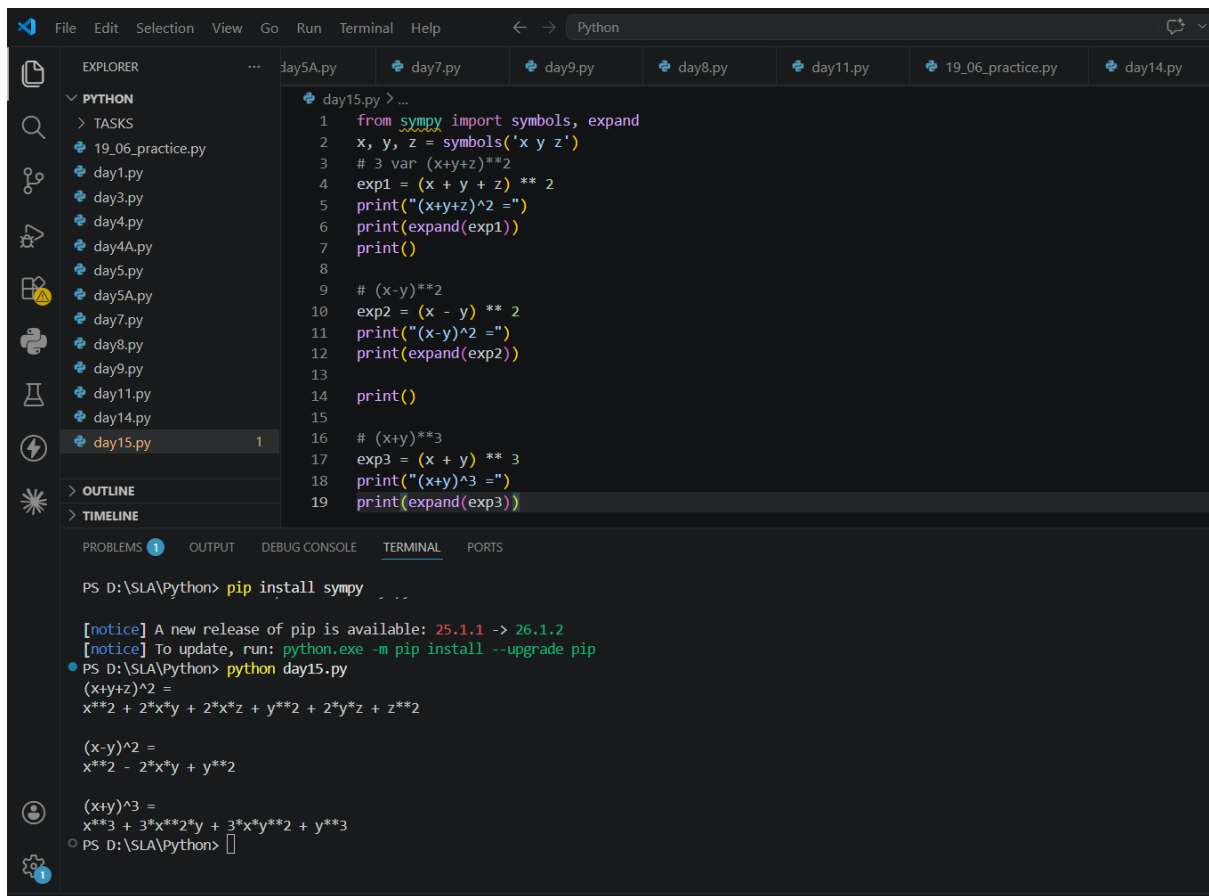


Day 15 Assignment



The image shows a Visual Studio Code editor with a Python file named `day15.py` open. The file contains a script that uses the SymPy library to perform symbolic expansions. The script defines three expressions: $(x+y+z)^2$, $(x-y)^2$, and $(x+y)^3$, and prints their expanded forms.

```
1 from sympy import symbols, expand
2 x, y, z = symbols('x y z')
3 # 3 var (x+y+z)**2
4 exp1 = (x + y + z) ** 2
5 print("(x+y+z)^2 =")
6 print(expand(exp1))
7 print()
8
9 # (x-y)**2
10 exp2 = (x - y) ** 2
11 print("(x-y)^2 =")
12 print(expand(exp2))
13
14 print()
15
16 # (x+y)**3
17 exp3 = (x + y) ** 3
18 print("(x+y)^3 =")
19 print(expand(exp3))
```

The terminal window at the bottom shows the command `pip install sympy` being executed, followed by a notice about a new release of pip. Then, the command `python day15.py` is executed, and the output of the script is displayed:

```
PS D:\SLA\Python> pip install sympy
[notice] A new release of pip is available: 25.1.1 -> 26.1.2
[notice] To update, run: python.exe -m pip install --upgrade pip
PS D:\SLA\Python> python day15.py
(x+y+z)^2 =
x**2 + 2*x*y + 2*x*z + y**2 + 2*y*z + z**2

(x-y)^2 =
x**2 - 2*x*y + y**2

(x+y)^3 =
x**3 + 3*x**2*y + 3*x*y**2 + y**3
PS D:\SLA\Python>
```